



Drought Monitoring with Python & GEE

Computing the Composite Drought Index (CDI)
from ERA5-Land and MODIS satellite data

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```
pip install drought-monitoring
```

Workshop Agenda

- 1 Why drought monitoring matters
- 2 The Composite Drought Index (CDI)
- 3 GEE + ERA5 + MODIS — data pipeline
- 4 Live coding walkthrough
- 5 Spatial maps & COG export
- 6 6-month forecast with STL + SARIMA
- 7 Q & A

5 min

15 min

20 min

35 min

15 min

10 min

10 min

Why Drought Monitoring Matters

2.5B

people live in dryland regions

\$300B

economic losses from drought
per decade globally

×3

drought frequency increase
in East Africa since 1990s

Kenya case study — Turkana County (2000–2025)

24/25 years below normal CDI · 2020 = worst on record (CDI 0.29) · 439,400 livestock deaths in 2022

Only 1 wet year (2005) in 25 years of satellite data

The Composite Drought Index — Formula

$$DI = (\mu_{IP} / \mu_{LTM}) \times \sqrt{ (RL_{IP} / RL_{LTM}) }$$

$$CDI = 0.50 \times PDI + 0.25 \times TDI + 0.25 \times VDI$$

μ_{IP}

Mean over 3-month interest period

RL_{IP}

Months inside window below (or above) LTM

μ_{LTM}

Long-term monthly mean (historical normal)

RL_{LTM}

Long-term mean of run lengths

Scale: $DI < 1$ = drought stress · $DI \approx 1$ = near normal · $DI > 1$ = wetter than normal

Three Drought Sub-Indices

PDI

Precipitation
Drought Index

Data source

50%

ERA5-Land
total_precipitation_sum

Anomaly rule

Deficit: $IP < LTM$
(below normal rainfall)

Rainfall is the primary drought driver

TDI

Temperature
Drought Index

Data source

25%

ERA5-Land
temperature_2m

Anomaly rule

Excess: $IP > LTM$
(above normal temperature)

Heat accelerates soil moisture loss

VDI

Vegetation
Drought Index

Data source

25%

MODIS/061/MOD13A3
NDVI band

Anomaly rule

Deficit: $IP < LTM$
(vegetation stress response)

Ecosystem integrated drought signal

CDI Severity Classification

< 0.50	Extreme drought
0.50 – 0.65	Severe drought
0.65 – 0.80	Moderate drought
0.80 – 0.90	Mild drought
0.90 – 1.10	Near normal
1.10 – 1.20	Mild wet
1.20 – 1.30	Moderately wet
> 1.30	Very wet

Turkana 2000–2025

2020	0.29	Worst year ever
2012	0.37	2nd worst
2013	0.38	3rd worst
2005	1.01	Only wet year
avg	0.80	25-yr mean

Installation & Setup

```
$ terminal
```

```
# Install the package (all extras)
pip install "drought-monitoring[all]"
```

```
# Or with uv (recommended)
uv add drought-monitoring
```

```
# Authenticate Google Earth Engine
earthengine authenticate # run once in terminal
```

```
# Python – first lines of every notebook
```

```
from drought_monitoring.gee \
    import authenticate
```

```
authenticate(
    project="your-gee-project"
)
```

```
# ✓ Ready to fetch data
```

[gee]

GEE / Earth Engine API

[cog]

COG raster export

[plot]

Matplotlib figures

[all]

Everything above

Step 1 — Fetch Data from Google Earth Engine

notebook.ipynb

```
aoi = [34.5, 1.5, 36.5, 5.5] # Turkana County [W, S, E, N]
```

```
from drought_monitoring.gee import (  
    fetch_era5_precip, # ERA5-Land · mm/month  
    fetch_era5_temp,   # ERA5-Land · °C  
    fetch_modis_ndvi,  # MODIS MOD13A3 · [-1, 1]  
)
```

```
precip = fetch_era5_precip(aoi, start_year=2000, end_year=2025)  
temp    = fetch_era5_temp(aoi, start_year=2000, end_year=2025)  
ndvi    = fetch_modis_ndvi(aoi, start_year=2000, end_year=2025)
```

```
# Each returns a pd.Series with a monthly DatetimeIndex  
print(precip.head()) # 2000-01-01    8.4 dtype: float64
```

ERA5-Land

ECMWF/ERA5_LAND/MONTHLY_AGGR

~9 km · 1950–present

Monitoring period

Min 20 years — Max 30 years

Default: 20 years

500 m · 2000–present

Step 2 — Compute CDI with compute_all()

```
from drought_monitoring import compute_all
```

```
df = compute_all(  
    precip,          # pd.Series  mm/month  
    temp,            # pd.Series  °C  
    ndvi,            # pd.Series  [-1, 1]  
    window = 3,      # 3-month interest period  
    weights = (0.50, 0.25, 0.25), # PDI, TDI, VDI  
)
```

```
# Returns a pd.DataFrame
```

```
print(df.tail(6))
```

```
#           PDI    TDI    VDI    CDI  
# 2025-07-01  0.74  0.91  0.68  0.77  
# 2025-08-01  0.82  0.88  0.71  0.81  
# 2025-09-01  0.89  0.95  0.75  0.87  
# 2025-10-01  0.61  1.02  0.58  0.70  
# 2025-11-01  0.43  1.08  0.49  0.61  
# 2025-12-01  0.38  1.12  0.41  0.57
```

PDI

Precipitation
Drought Index

50% weight

TDI

Temperature
Drought Index

25% weight

VDI

Vegetation
Drought Index

25% weight

CDI

Composite
Drought Index

weighted avg

Step 3 — Publication-Quality Plots

```
from drought_monitoring.plot import (
    plot_timeseries,      # CDI + sub-indices time series
    plot_anomaly_bars,    # Annual anomaly bar chart
    plot_seasonal_cycle,  # Monthly climatology  $\pm 1\sigma/\pm 2\sigma$ 
    plot_hovmoller,       # Latitude-time diagram
    plot_map_snapshot,    # Single-date spatial map
)

# Main time series figure
fig = plot_timeseries(
    df,
    title      = "CDI – Turkana County",
    subtitle   = "ERA5-Land + MODIS | 2000-2025",
)
fig.savefig("CDI_timeseries.png", dpi=300, bbox_inches="tight")

# Annual bar chart
fig2 = plot_anomaly_bars(df, freq="Y")
fig2.savefig("CDI_annual.png", dpi=300)
```

plot_timeseries

CDI + severity stripe
+ 12-month mean

plot_anomaly_bars

Annual/seasonal
anomaly bars

plot_seasonal_cycle

Monthly climatology
 $\pm 1\sigma / \pm 2\sigma$ shading

plot_hovmoller

Latitude–time
Hovmöller diagram

Step 4 — Spatial Maps & Cloud Optimized GeoTIFF Export

```
# Compute annual CDI maps (dask-parallel, no disk write)
ds = yearly_drought_maps(aoi, start_year=2000, end_year=2025)
# xr.Dataset — dims: (year, latitude, longitude)
# variables: PDI, TDI, VDI, CDI

# Plot all years
ds["CDI"].plot(
    col="time", col_wrap=4,
    cmap="RdBu", robust=True, figsize=(20,14)
)

# Export to Cloud Optimized GeoTIFFs
paths = cdi_stack_to_cog(
    ds, output_dir="outputs/",
    prefix="Turkana_2000_2025"
)

# outputs/Turkana_2000_2025_CDI.tif ← 1 band per year
```

COG Output Benefits

One band per year

YYYY date string per band

Range-request ready

Stream from S3 / GCS / Azure

leafmap compatible

No pre-processing needed

QGIS / ArcGIS ready

Standard GeoTIFF format

DEFLATE compressed

512×512 tile blocks

Step 5 — 6-Month Drought Forecast

```
from drought_monitoring.forecast import forecast_all_statistical
from drought_monitoring.plot import plot_forecast

# Fully statistical – no CDS API required
fc = forecast_all_statistical(
    precip, temp, ndvi,
    n_months = 6,          # 6-month horizon
    weights = (0.50, 0.25, 0.25),
    ci_level = 0.90,       # 90% confidence bands
)

# fc columns: PDI, TDI, VDI, CDI
#             PDI_lower, PDI_upper (bootstrap) CI)
#             TDI_lower, TDI_upper
#             VDI_lower, VDI_upper
#             CDI_lower, CDI_upper
#             lead (1 ... 6)

# Forecast plot with confidence fan
```

Forecast Methods

STL

Seasonal-Trend Decomposition

Extracts trend + seasonal components

SARIMA

SARIMA(1,0,0) (1,0,0)[12]

Models seasonal remainder

Bootstrap

300 draws from historical residuals

90% confidence band

VDI note

Raw NDVI → forecast → VDI

Preserves LTM calibration

Hands-On Exercise

Choose your own area of interest in Kenya and compute the CDI

1

Pick a county AOI

Nairobi: [36.6, -1.4, 37.1, -1.1]

Mombasa: [39.5, -4.2, 39.8, -3.9]

Nakuru: [35.8, -0.5, 36.3, 0.2]

2

Fetch 20 years of data

```
fetch_era5_precip, fetch_era5_temp, fetch_modis_ndvi  
(2000 - 2020)
```

3

Compute CDI

```
df = compute_all(precip, temp, ndvi)  
Identify your 3 worst drought years
```

4

Plot & compare

```
plot_timeseries(df) + plot_anomaly_bars(df)  
Where does your county rank vs Turkana?
```

5

Bonus: forecast

```
fc = forecast_all_statistical(precip, temp, ndvi, n_months=6)  
plot_forecast(df, fc)
```

What You've Learned Today

- ✓ The CDI formula: $(\mu_{IP} / \mu_{LTM}) \times \sqrt{(RL_{IP} / RL_{LTM})}$
- ✓ Why drought < 1.0 and wet > 1.0 on the CDI scale
- ✓ Fetching ERA5 + MODIS data directly from Google Earth Engine
- ✓ Computing PDI, TDI, VDI, and CDI with `compute_all()`
- ✓ Generating publication-quality climate science plots
- ✓ Exporting spatial results as Cloud Optimized GeoTIFFs
- ✓ Running a 6-month statistical forecast with STL + SARIMA